



Dr. Prateek Raj Gautam

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PROFESSIONAL SUMMARY

Assistant Professor with over **8 years** of teaching and research experience. **Ph.D.** in Electronics and Communication Engineering with interdisciplinary expertise in **Artificial Intelligence, Machine Learning, Reinforcement Learning, AIoT, Embedded Systems, Wireless Sensor Networks, Python, DevOps, Cloud Technologies**, and modern software engineering practices. Experienced in curriculum development, research supervision, cloud infrastructure, and project-based learning.

WORK EXPERIENCE

Assistant Professor – Senior Scale, School of Computer Science *February 2024 – Present*
UPES Dehradun, Uttarakhand.
Assistant Professor, Computer Science & Engineering *August 2022 – February 2024*
Centre for Advanced Studies, AKTU, Lucknow, UP.
Assistant Professor, Electronics & Communication Engineering *July 2013 – December 2015*
Allehnouse Institute of Technology, Kanpur, UP.
Assistant Professor, Electronics & Communication Engineering *June 2012 – July 2013*
Naraina College of Engineering and Technology, Kanpur, UP.

EDUCATION

Ph. D. *2016–2021*
Electronics & Communication Engineering (Wireless Sensor Networks), **Motilal Nehru National Institute of Technology Allahabad, Prayagraj (UP), India.** Thesis Title: “Energy Efficient 2D and 3D Localization in Wireless Sensor Networks using Single Anchor Node”.

M. Tech. *2009–2011*
Electronics & Communication Engineering, **Harcourt Butler Technological Institute (HBTI) Kanpur (UP), India**, with an aggregate of 67.55%. Thesis Title: “Generalized One Dimensional Optical Orthogonal Coding Scheme for CDMA Systems with its Grouping and Performance Analysis”.

B. Tech. *2004–2008*
Electronics & Communication Engineering, **University Institute of Engineering and Technology (UIET), CSJMU Kanpur (UP), India**, with an aggregate of 62.00%.

RESEARCH INTERESTS & COURSES DELIVERED

Wireless Sensor Networks (**WSNs**) / Internet of Things (**IoTs**), Energy efficient WSN Localization, Wireless Communication, **CDMA, IDMA**, Brain Wave Mapping. Artificial Intelligence (**AI**), Machine Learning (**ML**), Reinforcement Learning (**RL**), Computer Vision, Artificial Intelligence of Things (**AIoT**), Wireless Sensor Networks (**WSNs**), Internet of Things (**IoT**), Edge Computing, Embedded Systems, and Intelligent Distributed Systems.

Post Grad. (M.Tech.): Machine Learning, Computer Vision, Reinforcement Learning, Internet of Things, Data Communication Networks, Research Methodology, Concepts of Electrical & Electronics Engineering

Under Grad. (B.Tech.): Digital Electronics, Microprocessors, EMFT, Digital Communication, Measurements, Analog Integrated Circuits, Soft Computing, Python, Web Technologies, Full-Stack Development, DevOps and Containerization.

COMPUTER SKILLS

- Programming: **Python, Go, JavaScript, Bash, HTML5, CSS3.**
- AI/ML: **Artificial Intelligence, Machine Learning, Reinforcement Learning, Computer Vision**, NumPy, Pandas, Matplotlib.
- DevOps & Cloud: **Git, Docker, Kubernetes, Jenkins, CI/CD, Ansible**, Linux, Nix, NixOS, Self-hosted infrastructure.
- Web & Application Development: Django, WordPress, Kivy, Eel, REST APIs.
- Embedded Systems & IoT: Arduino IDE, PlatformIO, ESP32/Arduino programming, Sensors & Actuators integration, KiCAD, CST Studio Suite.
- Scientific Computing: MATLAB (previous collaborations: github.com/mgeekmatlab), LabVIEW, LTspice, Gnuplot.

• Technical Documentation & Visualization: L^AT_EX(TikZ, PGFPlots, Beamer), Inkscape, GIMP, Photoshop/CorelDraw, Blender.

SELECTED PROJECTS

Representative software and educational projects (additional projects available at projects.mgeek.in).

- **FormHelper** – Speed up data entry and form filling process.
- **GeneratorJS / SevnJS** – JavaScript libraries for frontend design
- **SlideGEN** – Selfhosted presentation generation from Markdown Content.
- Designed and deployed web applications using **Python, Django, HTML, CSS, JavaScript**, and modern DevOps workflows.
- Experience with containerized application deployment and management using **Docker, Kubernetes, Git, CI/CD pipelines, and Linux-based cloud infrastructure**.

PUBLICATIONS JOURNAL(J) CONFERENCE(C)

- [J1] **P. R. Gautam**, S. Kumar, A. Verma, T. Rashid, et al., “Energy-efficient localization of sensor nodes in WSNs using beacons from rotating directional antenna,” *IEEE Transactions on Industrial Informatics*, vol. 15, no. 11, pp. 5827–5836, Nov. 2019. DOI: [10.1109/tii.2019.2908437](https://doi.org/10.1109/tii.2019.2908437) issn [1551-3203](https://doi.org/10.1109/tii.2019.2908437) **Impact Factor: 12.3** **SCIE, Q1**
- [J2] **P. R. Gautam**, S. Kumar, A. Verma, and A. Kumar, “Energy-efficient localization of sensor nodes in wsns using single beacon node,” *IET Communications*, vol. 14, no. 9, pp. 1459–1466, 2020. DOI: [10.1049/iet-com.2019.1298](https://doi.org/10.1049/iet-com.2019.1298) issn [1751-8628](https://doi.org/10.1049/iet-com.2019.1298) **Impact Factor: 1.6** **SCIE, Q2**
- [J3] A. Verma, S. Kumar, **P. R. Gautam**, and A. Kumar, “Fuzzy logic based effective clustering of homogeneous wireless sensor networks for mobile sink,” *IEEE Sensors Journal*, vol. 20, no. 10, pp. 5615–5623, May 2020. DOI: [10.1109/jsen.2020.2969697](https://doi.org/10.1109/jsen.2020.2969697) issn [1530-437X](https://doi.org/10.1109/jsen.2020.2969697) **Impact Factor: 4.3** **SCIE, Q1**
- [J4] A. Verma, S. Kumar, **P. R. Gautam**, and A. Kumar, “Neural-fuzzy based effective clustering for large-scale wireless sensor networks with mobile sink,” *Peer-to-Peer Networking and Applications*, Jun. 2021. DOI: [10.1007/s12083-021-01167-6](https://doi.org/10.1007/s12083-021-01167-6) issn [1936-6450](https://doi.org/10.1007/s12083-021-01167-6) **Impact Factor: 4.2** **SCIE, Q2**
- [J5] A. Verma, S. Kumar, **P. R. Gautam**, T. Rashid, et al., “Broadcast and reliable coverage based efficient recursive routing in large-scale wsns,” *Telecommunication Systems*, vol. 75, no. 1, pp. 63–78, Jun. 2020. DOI: [10.1007/s11235-020-00679-5](https://doi.org/10.1007/s11235-020-00679-5) issn [1572-9451](https://doi.org/10.1007/s11235-020-00679-5) **Impact Factor: 2.5** **SCIE, Q2**
- [J6] M. Yadav, **P. R. Gautam**, V. Shokeen, and P. K. Singhal, “Modern fisher–yates shuffling based random interleaver design for SCFDMA-IDMA systems,” *Wireless Personal Communications*, vol. 97, no. 1, pp. 63–73, May 2017. DOI: [10.1007/s11277-017-4492-9](https://doi.org/10.1007/s11277-017-4492-9) issn [0929-6212](https://doi.org/10.1007/s11277-017-4492-9) **Impact Factor: 2.2** **SCIE, Q2**
- [J7] A. Verma, T. Rashid, **P. R. Gautam**, S. Kumar, et al., “Cost and sub-epoch based stable energy-efficient clustering algorithm for heterogeneous wireless sensor networks,” *Wireless Personal Communications*, vol. 107, no. 4, pp. 1865–1879, Apr. 2019. DOI: [10.1007/s11277-019-06362-6](https://doi.org/10.1007/s11277-019-06362-6) issn [0929-6212](https://doi.org/10.1007/s11277-019-06362-6) **Impact Factor: 2.2** **SCIE, Q2**
- [J8] T. Rashid, S. Kumar, A. Verma, **P. R. Gautam**, et al., “Co-reerp: Cooperative reliable and energy efficient routing protocol for intra body sensor network (intra-wbsn),” *Wireless Personal Communications*, vol. 114, no. 2, pp. 927–948, Apr. 2020. DOI: [10.1007/s11277-020-07401-3](https://doi.org/10.1007/s11277-020-07401-3) issn [0929-6212](https://doi.org/10.1007/s11277-020-07401-3) **Impact Factor: 2.2** **SCIE, Q2**
- [J9] S. Kumar, **P. R. Gautam**, A. Verma, T. Rashid, et al., “An energy-efficient transmission in wsns for different climatic conditions,” *Wireless Personal Communications*, vol. 110, no. 1, pp. 423–444, Sep. 2019. DOI: [10.1007/s11277-019-06735-x](https://doi.org/10.1007/s11277-019-06735-x) issn [0929-6212](https://doi.org/10.1007/s11277-019-06735-x) **Impact Factor: 2.2** **SCIE, Q2**
- [J10] S. Kumar, **P. R. Gautam**, T. Rashid, A. Verma, et al., “Division algorithm based energy-efficient routing in wireless sensor networks,” *Wireless Personal Communications*, Aug. 2021. DOI: [10.1007/s11277-021-08996-x](https://doi.org/10.1007/s11277-021-08996-x) issn [1572-834X](https://doi.org/10.1007/s11277-021-08996-x) **Impact Factor: 2.2** **SCIE, Q2**
- [J11] R. C. S. Chauhan, A. Kumar, and **P. R. Gautam**, “Optical orthogonal code generation scheme and grouping of codes for optical CDMA systems,” *International Journal of System Assurance Engineering and Management*, vol. 12, no. 1, pp. 91–103, 1 Jun. 2020. DOI: [10.1007/s13198-020-01007-5](https://doi.org/10.1007/s13198-020-01007-5) issn [0976-4348](https://doi.org/10.1007/s13198-020-01007-5) **Impact Factor: 2** **SCIE, Q3**
- [J12] **P. R. Gautam**, A. Verma, S. Kumar, D. Prasad, et al., “Design of directional antennas for wireless sensor networks and the internet of things experiments,” *IEEE Sensors Letters*, vol. 6, no. 9, pp. 1–4, 2022. DOI: [10.1109/LSSENS.2022.3202919](https://doi.org/10.1109/LSSENS.2022.3202919) issn [2475-1472](https://doi.org/10.1109/LSSENS.2022.3202919) **Impact Factor: 2.8** **SCIE, Q2**

- [J13] Shilpi, **P. R. Gautam**, S. Kumar, and A. Kumar, "An optimized sensor node localization approach for wireless sensor networks using rssi," *The Journal of Supercomputing*, vol. 79, pp. 7692–7716, 2022. DOI: <https://doi.org/10.1007/s11227-022-04971-w> issn 0920-8542 **Impact Factor: 3.3** **SCIE, Q2**
- [J14] A. Verma, S. Kumar, **P. R. Gautam**, T. Rashid, et al., "Enhanced cost and sub-epoch based stable energy-efficient clustering algorithm for heterogeneous wireless sensor networks," *Wireless Personal Communications*, Jul. 2023. DOI: [10.1007/s11277-023-10601-2](https://doi.org/10.1007/s11277-023-10601-2) issn 1572-834X **Impact Factor: 2.2** **SCIE, Q2**
- [J15] S. Kumar, **P. R. Gautam**, T. Rashid, A. Verma, et al., "ETDCC: Energy-efficient transmission scheme for dynamic climatic conditions in WSN," *TELKOMNIKA (Telecommunication Computing Electronics and Control)*, vol. 16, no. 3, p. 1126, Jun. 2018. DOI: [10.12928/telkomnika.v16i3.8513](https://doi.org/10.12928/telkomnika.v16i3.8513) issn 1693-6930 **Scopus**
- [J16] T. Rashid, S. Kumar, A. Verma, **P. R. Gautam**, et al., "Pm-EEMRP: Postural movement based energy efficient multi-hop routing protocol for intra wireless body sensor network (intra-WBSN)," *TELKOMNIKA (Telecommunication Computing Electronics and Control)*, vol. 16, no. 1, p. 166, Feb. 2018. DOI: [10.12928/telkomnika.v16i1.7318](https://doi.org/10.12928/telkomnika.v16i1.7318) issn 1693-6930 **Scopus**
- [J17] A. Verma, T. Rashid, **P. R. Gautam**, S. Kumar, et al., "Fuzzy based stable clustering protocol for Heterogeneous wireless sensor networks," *International Journal of Engineering and Technology*, vol. 9, no. 4, pp. 2854–2860, Aug. 2017. DOI: [10.21817/ijet/2017/v9i4/170904046](https://doi.org/10.21817/ijet/2017/v9i4/170904046) issn 0975-4024 **Scopus 2017**
- [J18] T. Rashid, S. Kumar, A. Verma, **P. R. Gautam**, et al., "RB-IEMRP: Relay based improved throughput energy-efficient multi-hop routing protocol for intra body sensor network (INTRA-WBSN)," *International Journal of Computer Networks & Communications*, vol. 11, no. 02, pp. 69–82, Mar. 2019. DOI: [10.5121/ijcnc.2019.11205](https://doi.org/10.5121/ijcnc.2019.11205) issn 0974-9322 **Scopus**
- [C1] **P. R. Gautam**, S. Kumar, A. Verma, and A. Kumar, "Localization of sensor nodes in WSNs using three dimensional angle of arrival detection at BS," in *2019 International Conference on Electrical, Electronics and Computer Engineering (UPCON)*, ZHCET, AMU, Aligarh: IEEE, Nov. 2019, pp. 1–4. DOI: [10.1109/upcon47278.2019.8980262](https://doi.org/10.1109/upcon47278.2019.8980262) isbn: 9781728134550 issn 2687-7767 **Scopus**
- [C2] **P. R. Gautam**, S. Kumar, A. Verma, T. Rashid, et al., *Localization of Sensor Nodes in WSN Using Area Between a Node and Two Beacons* (Lecture Notes in Electrical Engineering). Motilal Nehru National Institute of Technology, Allahabad: Springer, Dec. 2019, vol. 587, pp. 221–228, 1060 pp. DOI: [10.1007/978-981-32-9775-3_22](https://doi.org/10.1007/978-981-32-9775-3_22) isbn: 9813297743 issn 1876-1100 **Book chapter**
- [C3] **P. R. Gautam**, S. Kumar, and A. Kumar, "Sensor localization in wsns using rotating directional - antenna at the base station," in *Advances in VLSI, Communication, and Signal Processing*, ser. Lecture Notes in Electrical Engineering, vol. 683, Motilal Nehru National Institute of Technology, Allahabad: Springer, Oct. 2020, pp. 705–718. DOI: [10.1007/978-981-15-6840-4_58](https://doi.org/10.1007/978-981-15-6840-4_58) isbn: 978-981-15-6839-8 issn 1876-1100 **Book chapter**
- [C4] A. Kumar, S. Kumar, **P. R. Gautam**, A. Verma, et al., *Performance Evaluation of Multi-operands Floating-Point Adder* (Lecture Notes in Electrical Engineering). JK Institute of Applied Physics and Technology, Allahabad University, Allahabad: Springer Singapore, Dec. 2019, vol. 524, pp. 537–546. DOI: [10.1007/978-981-13-2685-1_51](https://doi.org/10.1007/978-981-13-2685-1_51) isbn: 9811326843 issn 1876-1119 **Book chapter**
- [C5] S. Kumar, A. Verma, **P. R. Gautam**, A. Dayal, et al., "The load balancing of optimizing LEACH clustering algorithm with mobile sink and rendezvous nodes," in *2018 5th IEEE Uttar Pradesh Section International Conference on Electrical, Electronics and Computer Engineering (UPCON)*, Madan Mohan Malaviya University of Technology, Gorakhpur: IEEE, Nov. 2018. DOI: [10.1109/upcon.2018.8596989](https://doi.org/10.1109/upcon.2018.8596989) isbn: 978-1-5386-5002-8 issn 2687-7759 **Scopus**
- [C6] S. Kumar, **P. R. Gautam**, A. Verma, R. Verma, et al., *Energy Efficient Routing using Sectors Based Energy-Hole Reduction in WSNs*. ZHCET, AMU, Aligarh: IEEE, 2019. DOI: [10.1109/upcon47278.2019.8980254](https://doi.org/10.1109/upcon47278.2019.8980254) isbn: 978-1-7281-3455-0 issn 2687-7767 **Scopus**
- [C7] A. Verma, S. Kumar, **P. R. Gautam**, and A. Kumar, *Stable Energy-Efficient Routing Algorithm for Dynamic Heterogeneous Wireless Sensor Networks* (Lecture Notes in Electrical Engineering). Motilal Nehru National Institute of Technology, Allahabad: Springer, Dec. 2019, vol. 587, pp. 221–228, 1060 pp. DOI: [10.1007/978-981-32-9775-3_15](https://doi.org/10.1007/978-981-32-9775-3_15) isbn: 9813297743 issn 1876-1100 **Book chapter**

- [C8] S. Shilpi, **P. R. Gautam**, S. Kumar, and A. Kumar, “A comparative analysis of distance-based node localization in wireless sensor network,” in *2021 8th International Conference on Signal Processing and Integrated Networks (SPIN)*, vol. 0, 2021, pp. 118–123. DOI: [10.1109/SPIN52536.2021.9566136](https://doi.org/10.1109/SPIN52536.2021.9566136) isbn: 9781665435642 issn 0 **Scopus**
- [C9] M. Yadav, **P. R. Gautam**, and K. Singhal P., “Inverse tree interleavers in uav communications for interference mitigation,” in *Decision Support Systems for Smart City Applications* (Concise Introductions to AI and Data Science), Concise Introductions to AI and Data Science. John Wiley & Sons, Ltd, Dec. 2022, ch. 3, pp. 35–52. DOI: [10.1002/9781119896951.ch3](https://doi.org/10.1002/9781119896951.ch3) isbn: 9781119896951 issn **Book chapter**
- [C10] A. Rukasar and P. R. Gautam, “Lane detection and tracking algorithms for driver assistance system,” in 2023, pp. 872–879. DOI: [10.1109/icac3n60023.2023.10541447](https://doi.org/10.1109/icac3n60023.2023.10541447) isbn: 9798350330861 issn **scopus**
- [C11] V. Kumar, A. Kumar, and **P. R. Gautam**, “Dental disease detection and classification in radiograph images using deep learning model,” in 2023, pp. 1198–1203. DOI: [10.1109/ICAC3N60023.2023.10541747](https://doi.org/10.1109/ICAC3N60023.2023.10541747) isbn: 9798350330861 issn **scopus**
- [C12] N. Awasthi, **P. R. Gautam**, and A. Sharma, “Rfecv-dt: Recursive feature selection with cross validation using decision tree based android malware detection,” in 2024. DOI: [10.1109/ICCCNT61001.2024.10725127](https://doi.org/10.1109/ICCCNT61001.2024.10725127) isbn: 9798350370249 issn **scopus**
- [C13] M. Ansari and **P. R. Gautam**, “Classification of soil moisture content with the application of deep learning,” in 2024. DOI: [10.1109/ICIC3S61846.2024.10603387](https://doi.org/10.1109/ICIC3S61846.2024.10603387) isbn: 9798350364088 issn **scopus**
- [C14] A. Sharma, A. Upadhyay, and **P. R. Gautam**, “Prediction of water discharge in mahanadi river basin, india using artificial neural networks,” in 2025, vol. 1, pp. 28–32. DOI: [10.1201/9781003501244-5](https://doi.org/10.1201/9781003501244-5) isbn: 9781032911571 issn **scopus**
- [C15] P. Mishra, J. Singh, and **P. R. Gautam**, “Mustard and wheat mildew disease classification using deep learning,” in 2025, vol. 1, pp. 72–78. DOI: [10.1201/9781003501244-14](https://doi.org/10.1201/9781003501244-14) isbn: 9781032911571 issn **scopus**
- [C16] A. Upadhyay, A. Sharma, and **P. R. Gautam**, “Estimation of sediment load in mahanadi river, india using artificial neural networks,” in 2025, vol. 1, pp. 23–27. DOI: [10.1201/9781003501244-4](https://doi.org/10.1201/9781003501244-4) isbn: 9781032911571 issn **scopus**
- [C17] M. Ansari and **P. R. Gautam**, “Comparison of different pre-trained deep learning models for classification of soil moisture content,” in 2025, vol. 2, pp. 107–113. DOI: [10.1201/9781003561651-15](https://doi.org/10.1201/9781003561651-15) isbn: 9781032911571 issn **scopus**

PAPER PRESENTED

“**Localization of Sensor Nodes in WSNs using Three Dimensional Angle of Arrival detection at BS**” In *1st International Conference on VLSI, Communication and Signal Processing (VCAS 2018)* at MNNIT Allahabad (UP) India. *29th November to 1st December 2018*

“**Sensor Localization in WSNs Using Rotating Directional - Antenna at the Base Station.**” In *2nd International Conference on VLSI, Communication and Signal Processing (VCAS 2019)* at MNNIT Allahabad (UP) India. *21st – 23rd October 2019*

JOURNAL REVIEWER AND EDITOR

• IEEE Transactions on Industrial Informatics WOS (3), • IET Communications WOS (6),
• International Journal of Distributed Sensor Networks WOS (1), • Asian Journal of Cardiology Research (1), • SN Applied Sciences WOS (1), • Telecommunication Systems (3), • Journal of Optical Communications (1), • Optica Applicata (1), • International Journal of Procurement Management (1), and • **AE IJSAEM (8)**.

WORKSHOPS /FDP

1. **One-week GIAN workshop “Advances in Nanotechnology and its Application in Future Electronics (ANFE-2017)” held at MNNIT Allahabad.**
– Participated and volunteered *6th – 10th November, 2017*
2. **Ten days GIAN workshop on “Internet of Things in Smart Living & Cyber-Physical-Social Systems” held at IIT Kanpur.**
– Participated and volunteered. *8th – 17th January 2018*
3. **Summer training program on “VLSI Design & Embedded System (VDES-2018)” held at MNNIT Allahabad.**
– Volunteered. *13th June – 12th July, 2018*
4. **ATAL Academy FDP on “Blockchain” held at MNNIT Allahabad.**
– Participated. *16th – 20th September 2019*

5. ATAL Academy FDP on “Artificial Intelligence” held at MNNIT Allahabad.
– Participated. *10th – 14th December 2019*
6. One-week short term course on “Computational Physics” held at MNNIT Allahabad.
– Participated. *1st – 5th March 2021*
7. One-week FDP on “IPR Awareness and Patent Prosecution” held at MNNIT Allahabad.
– Participated. *13th – 17th July 2021*
8. One-week FDP on “Antenna Design and Microwave Applications” held at HBTU Kanpur.
– Participated. *23th – 27th July 2021*

WORKSHOPS FACILITATED

Manuscript preparation in LaTeX,
Programming with 8051 micro-controller.

AWARDS AND OTHER ACHIEVEMENTS

1. Awarded national scholarship “RG-NFSC” from UGC. *2017-2021*
 2. Offered national scholarship “MANF” from UGC based on NET score *2020*
 3. Eight times GATE qualified. *2008, 2009, 2012, 2013, 2014, 2016, 2017, and 2020*
 4. Three times UGC NET (Electronics Science) qualified. *Jul-2016, Jan-2017, and Dec-2019*
 5. Member of IEEE, IEEE Industrial Electronics Society, IEEE Microwave Theory and Techniques Society, and IEEE Broadcast Technology Society.
 6. NPTEL Online Certification on MATLAB for Numerical Computations.
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REFERENCES

Available on demand.

Prateek Raj Gautam

Prateek Raj Gautam
June 6, 2026
